

LLM Judge API Budget — MAM-AI v0.2 HealthBench

Two budget items for the HealthBench rubric track of the v0.2 evaluation, both via the closed-source Batch API. *(Separate smaller asks of \$30 each cover the SAQ rescore and the faithfulness judge.)*

Phase	Purpose	Estimated batch cost	Ask
A. Calibration	Validate gpt-5.5 as LLM judge against physician ground truth	\$96	\$120
B. Production rescore	Score 38,308 model responses on the HealthBench rubric track	\$533	\$600

Current ask total: \$720. A further ~\$600 covers 2 anticipated retrieval-module iterations (see Forward-looking section) — filed separately when each triggers, not part of this approval. Full project envelope across v0.2 + 2 iterations: **~\$1,320.**

Phase A validates **gpt-5.5** as the judge against physician ground truth before Phase B uses it on production data. Both phases are required: the open-source bake-off (see Background) showed that open-weight judges over-rate "met" on a meaningful fraction of physician-not-met rows — biasing the deployment-headline rubric metric in the wrong direction — so a frontier closed-source judge is necessary for the final v0.2 evaluation.

Background

MAM-AI v0.2's open-ended evaluation tracks use an LLM judge to score Gemma 4 E4B responses against physician-written rubric criteria. We **already** ran an open-source bake-off across three candidates (Llama 4 Maverick FP8, Nemotron-Ultra-253B FP8-quantized, gpt-oss-120b) on a 6,853-row physician-labeled meta-evaluation set ([mamabench@v0.2.1/calibration/obgyn_meta_eval.jsonl](#)). Result: **all three open candidates systematically over-rate "met"** on the rows where physicians unanimously agreed the criterion was **NOT** met.

Judge (open-weight)	Footprint	Judge met-rate vs phys 77%	Not-met agreement	Over-rated rate on not-met rows
Llama 4 Maverick FP8	4× H200	88.8% (+11.8 pp)	44.6%	55.4%
Nemotron-Ultra-253B (FP8 q.)	4× H200	83.3% (+6.3 pp)	55.3%	44.7%
gpt-oss-120b (medium, pinned)	1× H200	73.2% (−3.9 pp)	67.9%	32.1%

The pinned candidate (gpt-oss-120b medium) is the **best** of the three. We also confirmed higher reasoning effort doesn't fix it: gpt-oss-120b at **reasoning_effort=high** improved not-met agreement

by only 1.7 pp (67.9% → 69.6%) while degrading other metrics — the task is not reasoning-bound. **Bigger ≠ better either**: the two heavier candidates score worse, not better.

A judge that over-rates inflates the rubric headline metric (**weighted_met**) — the exact direction we don't want for a deployment story. We've exhausted the obvious open-source levers (3 models, 2 reasoning settings on the best one) and the bias persists across all of them. **The final evaluation requires a frontier closed-source judge**: we need to calibrate **gpt-5.5** against the same physician-labeled set (Phase A), confirm it materially improves the not-met agreement, and then run it over the production rubric track (Phase B).

Phase A — Calibration (**gpt-5.5** vs physician ground truth)

Task

Run **gpt-5.5** over the OBGYN-scoped HealthBench grader meta-evaluation set (**mamabench@v0.2.1/calibration/obgyn_meta_eval.jsonl**) — **6,853 physician-labeled (prompt, completion, criterion) triples**. Compute a 4-metric report: LLM↔single-human agreement, LLM↔consensus, judge met-rate vs physician met-rate, and per-class agreement (rubber-stamp detector).

Why necessary

The Background section establishes that open-weight judges over-rate "met" on physician-not-met rows. Before committing the Phase B spend, Phase A validates that **gpt-5.5** is the upgrade we expect — by running it over the same 6,853 physician-labeled triples and producing a side-by-side metric report against the open-weight bake-off. The output is a precondition for trusting any production Phase B verdicts.

Cost calculation

Pricing: **gpt-5.5** Batch = \$2.50 / 1M input · \$15 / 1M output (50% off the standard \$5 / \$30).

Token volume (measured from actual calibration data):

	per call	× 6,853 calls
Input (grader template + conversation + completion + criterion)	~1,361	9.33 M
Output at reasoning_effort=medium	~700	4.80 M
Component	Batch cost	
Input: 9.33 M × \$2.50/M	\$23.33	
Output: 4.80 M × \$15/M	\$72.00	
Estimated total	\$96	
Ask (with ~25% buffer for output-token overruns)	\$120	

Phase B — Production Rescore (HealthBench rubric, ±RAG)

Task

For each (model response, criterion) pair across the 3 HealthBench configs and both ±RAG arms, ask the pinned judge whether the criterion is met. The HealthBench `weighted_met` headline metric is computed per response from these binary verdicts.

Why necessary

This **is** the published HealthBench evaluation methodology. Without it, the v0.2 report cannot:

- Quantify Gemma's clinical-rubric quality (the deployment-relevant metric).
- Compare ±RAG on the rubric track (key for the RAG deployment claim).
- Compare against the published HealthBench leaderboard.

A frontier closed-source judge is required because the open-weight bake-off (see Background) showed every open candidate biases the headline metric upward by over-rating "met." The validated `gpt-5.5` judge from Phase A is the one that goes into production here.

Cost calculation

Call count (one judge call per criterion, matching HealthBench's published methodology):

Config	rows × avg criteria/row	criterion calls / arm
healthbench_oss_eval	1,209 × 11.7	14,195
healthbench_consensus	872 × 2.2	1,909
healthbench_hard	258 × 11.8	3,050
Per arm		19,154
Total × 2 arms (±RAG)		38,308

Token volume:

	per call	× 38,308 calls
Input	~1,361	52.14 M
Output at <code>reasoning_effort=medium</code>	~700	26.82 M
Component		Batch cost
Input: 52.14 M × \$2.50/M		\$130.35
Output: 26.82 M × \$15/M		\$402.30
Estimated total		\$533
Ask (with ~13% buffer for output-token variance)		\$600

Forward-looking estimate — future retrieval-iteration runs

The asks above cover **one full v0.2 evaluation of the current MAM-AI app**. Over the coming months, planned retrieval-module improvements will trigger re-runs of the **+RAG arm only** — the no-RAG baseline doesn't change because it doesn't use retrieval.

Each retrieval improvement re-runs roughly *half* of Phase B (one arm of 19,154 criterion calls instead of both arms' 38,308), plus a small SAQ re-run:

Per improvement	Calls	Batch cost
HealthBench rubric, +RAG arm only	19,154	~\$267
SAQ, +RAG arm only	369	~\$10
Per-improvement total		~\$277

Anticipating **2 retrieval-module improvements** over the coming months:

	Cost
Per improvement	~\$277
× 2 improvements	~\$554
Forward-looking estimate (with ~10% buffer)	~\$600

This is not part of the current ask — each improvement would be a separately approved follow-up. It is provided so the supervisor has visibility into the full retrieval-iteration roadmap: ~\$720 for the current v0.2 evaluation, then ~\$300 per retrieval improvement.

Total budget across closed-source judge candidates

Token volumes are fixed regardless of vendor or tier (the workload is the same); only price-per-token differs. **Reference token volumes** for all calculations below:

Phase	Calls	Input tokens	Output tokens (at <i>reasoning_effort=medium</i>)
A — Calibration (one-time)	6,853	9.33 M	4.80 M
B — Production rescore (one-time, both ±RAG arms)	38,308	52.14 M	26.82 M
Per future retrieval iteration (+RAG arm only, HealthBench + SAQ)	19,523	27.55 M	13.78 M

Per-candidate cost — pricing + project total (1× Phase A + 1× Phase B + 2× future iterations)

Grouped by vendor (OpenAI → Anthropic → Google). Prices per 1 M tokens. **All three vendors offer a Batch API at 50% off standard — the prices shown are those batch rates.**

Vendor / Tier	Input \$/M	Output \$/M	Project total	+~12% buffer
OpenAI gpt-5.5	\$2.50	\$15.00	\$1,181	~\$1,320
OpenAI gpt-5.4	\$1.25	\$7.50	\$590	~\$660
OpenAI gpt-5	\$0.625	\$5.00	\$369	~\$420
Claude Opus 4.7	\$2.50	\$12.50	\$1,031	~\$1,160
Claude Sonnet 4.6	\$1.50	\$7.50	\$618	~\$700
Gemini 3.1 Pro Preview	\$1.00	\$6.00	\$471	~\$530

Recommendation

Three paths, depending on how aggressively cost is a binding constraint.

Path 1 — No compromise on capability. Pick either:

- **OpenAI gpt-5.5** (~\$1,320 with buffer) — OpenAI's top-tier reasoning model.
- **Claude Opus 4.7** (~\$1,160 with buffer) — Anthropic's frontier. Comparable capability, slightly cheaper.

The asks in earlier sections of this doc (\$120 + \$600 + \$600) anchor on gpt-5.5.

Path 2 — Keep OpenAI recognition, keep cost low. Pick **OpenAI gpt-5** (~\$420 with buffer). Still a recognized OpenAI flagship — the methods section reads "OpenAI's reasoning model" — at roughly a **third** the cost of gpt-5.5. Tradeoff: weaker reasoning capacity than 5.5/5.4, so the "frontier" defense in the methods section is softer.

Path 3 — Save every penny. Pick **Gemini 3.1 Pro Preview**. Frontier-class capability from Google. API charges would be ~\$530 with buffer, but **new Google Cloud accounts receive \$300 free credits each on sign-up** (credit-card-linked) — stacking 2 fresh accounts covers the full spend, **net \$0 out-of-pocket**. Tradeoff is logistical, not financial: requires onboarding multiple fresh Google Cloud accounts with separate credit cards.

The mid-tier options (gpt-5.4, Sonnet 4.6) sit between Paths 1 and 2 and can be considered if none of the three fit.